Project: Worked examples
Title: Containment fill, bend radius and cable pulling
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the fill of a conduit, trunking or tray, the bend radius and the pull of the cables through a route. The space factor, bend multiple, friction coefficient and pulling limits are your own records.

WHAT IT CALCULATES	RELATION
Cable area	$A_{cab} = \Sigma\, n_i \cdot \pi \cdot d_i^2/4$ (n_i cables of overall diameter d_i)
Containment fill	$A_{int} = \pi \cdot D^2/4$ (conduit, internal diameter D) or W·H (trunking or tray, internal width × depth); fill = A_{cab}/A_{int}; u_{fill} = fill/f_{max}, f_{max} the space factor from your record
Bend radius	$R_{min} = n_{bend} \cdot d_{max}$, d_{max} the largest cable diameter, n_{bend} from your record; each bend $u_b = R_{min}/R$
Cable weight	$w = \Sigma\, n_i \cdot m_i$ (kg/m); force per metre $w \cdot g \cdot w_c$, g = 9.80665 m/s², w_c your weight-correction factor (1 = full weight)
Straight length	$T_2 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
Tension at a bend	$T_2 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$, $\mu_e = w_c \cdot \mu$, θ in radians (horizontal bend, cable on the containment floor, the default); capstan option, no weight term: $T_2 = T_1 \cdot e^{(\mu_e \cdot \theta)}$
Sidewall pressure and direction	SWP = T_2/R; the route is run in the listed order and in reverse, and the direction that passes every tension and sidewall pressure check is recommended (else the one of lower maximum tension)
Pulling limits	$T_{cab} = k_n \cdot N \cdot \min(T_{max,i})$ (load shared equally), u_T = T/T_{cab}; $u_{grip} = T/T_{grip}$, T_{grip} your grip or pulling-eye limit; $u_P = SWP_{max}/P_{lim}$

Basis: Other mechanical stresses on wiring systems (damage during installation, bending radius)[1](reg. 522.8); Conduit systems for cable management: general requirements (edition to be checked)[2]; Friction: belt friction (the capstan equation)[3](Ch. 8, 11th ed. (edition to be checked)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
cables	Cables in the containment and pulled together	2 row(s)		spec: Cable data sheets CDS-21 (servo power) and CDS-22 (encoder): overall diameter, mass and maximum pulling tension	any
$cont_{type}$	Containment	Conduit (round)		drawing: CND-ROUTE-07 rev C	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$bend_{model}$	Tension round a bend: horizontal bend with the cable's weight on the containment floor (default), or the capstan equation without the weight term	Horizontal bend, cable on the containment floor (weight term)		default	any
route	Pull route in pull order (straight lengths and bends)	6 row(s)		drawing: CND-ROUTE-07 rev C	any
D_{int}	Conduit internal diameter (conduit)	40	mm	drawing: CND-ROUTE-07 rev C	10 … 300 mm
T_1	Tension at the start of the pull (pulling rope, drum or head)	150	N	spec: Pulling method statement PMS-4: 150 N rope and pulling-head tension at the start of the pull	0 … 10000 N
μ	Coefficient of friction, cable sheath on containment (from your record)	0.3		reference: record: coefficient of friction, sheath on lubricated conduit (a value invented for this example; enter your own)	0.05 … 1
f_{max}	Space factor limit (fill limit, from your record)	0.5		reference: record: space factor for this arrangement (a value invented for this example; enter your own from your installation standard or company rule)	0.1 … 0.6
n_{bend}	Minimum bend radius as a multiple of the largest cable diameter (from your record)	10		reference: record: minimum bend radius multiple (a value invented for this example; enter the largest multiple from the cable data sheets)	3 … 30

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
P_{lim}	Sidewall pressure limit, force per metre of bend (from the cable maker)	3000	N/m	reference: record: sidewall pressure limit (a value invented for this example; enter the cable maker's figure)	100 ... 100000 N/m

TABLE ROW	VALUES
cables[1]	name = Servo power; od = 14 mm; count = 2; mass = 0.35 kg/m; T_max = 600 N
cables[2]	name = Encoder and control; od = 9 mm; count = 4; mass = 0.12 kg/m; T_max = 200 N
route[1]	kind = Straight; L = 6 m
route[2]	kind = Bend; theta = 90°; R = 500 mm
route[3]	kind = Straight; L = 14 m
route[4]	kind = Bend; theta = 90°; R = 400 mm
route[5]	kind = Straight; L = 9 m
route[6]	kind = Bend; theta = 45°; R = 600 mm

3. Assumptions

A1. Cables are circular and counted by overall diameter. (Fill by geometry; the factor is a user record.)
A2. There is no packing or cable and conduit factor method. (Fill by geometry; the factor is a user record.)
A3. The jam-ratio warning is the only jamming check. (Fill by geometry; the factor is a user record.)
A4. The space factor is yours, from your own record. (Fill by geometry; the factor is a user record.)
A5. Every cable is pulled together as one bundle on one pulling head with the load shared equally. (Equal sharing; unequal sharing is not modelled.)
A6. Each cable carries T/N, so the weakest cable governs. (Equal sharing; unequal sharing is not modelled.)
A7. The cable limit is N times the lowest maximum pulling tension, derated by your factor above 3 cables. (Equal sharing; unequal sharing is not modelled.)
A8. The grip or pulling-eye limit is a separate limit you enter. (Equal sharing; unequal sharing is not modelled.)
A9. Straight lengths are horizontal and the cable weight bears on the containment over the full length. (Equivalent friction $w_c \cdot \mu$ at the bends; blank w_c means 1.)
A10. Bends are taken in the plane of the route. (Equivalent friction $w_c \cdot \mu$ at the bends; blank w_c means 1.)
A11. Inclines, vertical runs and pulley or roller reductions are not modelled, so fold them into μ. (Equivalent friction $w_c \cdot \mu$ at the bends; blank w_c means 1.)
A12. A cradled or triangular weight correction is entered as w_c. (Equivalent friction $w_c \cdot \mu$ at the bends; blank w_c means 1.)
A13. The factor w_c scales the weight on straight lengths and the friction in bends. (Equivalent friction $w_c \cdot \mu$ at the bends; blank w_c means 1.)
A14. The contact force of a cradled or triangular arrangement is w_c times that of a flat one. (Equivalent friction $w_c \cdot \mu$ at the bends; blank w_c means 1.)
A15. On a straight length $T_2 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$. (Equivalent friction $w_c \cdot \mu$ at the bends; blank w_c means 1.)
A16. At a bend $\mu_e = w_c \cdot \mu$ in the cosh/sinh or capstan form. (Equivalent friction $w_c \cdot \mu$ at the bends; blank w_c means 1.)
A17. A bend in a vertical plane has the weight along the cable rather than on the floor. (Equivalent friction $w_c \cdot \mu$ at the bends; blank w_c means 1.)
A18. For that bend, take the capstan option and fold the rise into the route. (Equivalent friction $w_c \cdot \mu$ at the

bends; blank w_c means 1.)
A19. The coefficient of friction is one constant for the whole route. (A user record with its source.)
A20. Static and running friction are not distinguished. (A user record with its source.)
A21. Sidewall pressure is the tension leaving the bend over the bend radius, for the bundle as a whole. (SWP = T_2/R.)
A22. Enter the cable maker's per-cable limit unchanged. (SWP = T_2/R.)
A23. The bundle T/R against it is conservative for a triangular arrangement and roughly equal for a cradled one. (SWP = T_2/R.)
A24. Do not multiply the limit by the number of cables. (SWP = T_2/R.)
A25. The bend radius uses the individual cable diameter. (Cable data sheet.)
A26. For a triplexed or bundled assembly, enter the assembly diameter from the cable maker as the overall diameter. (Cable data sheet.)
A27. The tension at the start of the pull is the same in both directions. (Weight term for a horizontal bend; capstan equation as the option.)
A28. With the horizontal-bend (weight) model a heavy cable gains tension at a bend even from $T_1 = 0$. (Weight term for a horizontal bend; capstan equation as the option.)
A29. With the capstan option a bend at the very start of a pull with no start tension adds nothing. (Weight term for a horizontal bend; capstan equation as the option.)
A30. Enter the pulling rope or drum tension when it matters. (Weight term for a horizontal bend; capstan equation as the option.)

4. Working

[1] $A_{c1} = n \cdot \pi \cdot d^2/4$
$= 2 \times \pi \times (14\ \text{mm})^2 / 4$
$= 307.88\ \text{mm}^2$
Servo power: cross-section of 2 cable(s)

[2] $A_{c2} = n \cdot \pi \cdot d^2/4$
$= 4 \times \pi \times (9\ \text{mm})^2 / 4$
$= 254.47\ \text{mm}^2$
Encoder and control: cross-section of 4 cable(s)

[3] $A_{cab} = \Sigma\ n_i \cdot \pi \cdot d_i^2/4$
$= A_{c1} + A_{c2}$
$= 562.35\ \text{mm}^2$
Total cable cross-section

[4] $OD_{max} = \max\ d_i$
$= 14\ \text{mm}$
$= 14\ \text{mm}$
Largest cable overall diameter

[5] $w = \Sigma\ n_i \cdot m_i$
$= 2 \times 0.35\ \text{kg/m} + 4 \times 0.12\ \text{kg/m}$
$= 1.18\ \text{kg/m}$
Mass per metre of the bundle

[8] $A_{int} = pi * D_{int}\text{^}2 / 4$
$= \pi \times (40\ \text{mm})\text{^}2 / 4$
$= 1256.6\ \text{mm}^2$
Internal cross-section of the conduit

[10] $fill = A_{cab} / A_{int}$
$= 562.35\ \text{mm}^2 / 1256.6\ \text{mm}^2$
$= 44.75\ \%$
Fill: cable cross-section over internal cross-section

[11] $u_{fill} = A_{cab} / (A_{int} * f_{max})$
$= 562.35\ \text{mm}^2 / (1256.6\ \text{mm}^2 \times 0.5)$
$= 0.895$
Fill utilisation against the space factor

[13] $R_{min} = n_{bend} * OD_{max}$
= 10 × 14 mm
= 140 mm
Minimum bend radius for the largest cable
[14] $u_{b2} = R_{min}/R$
= 140 mm / 500 mm
= 0.28
Segment 2: bend radius utilisation
[15] $u_{b4} = R_{min}/R$
= 140 mm / 400 mm
= 0.35
Segment 4: bend radius utilisation
[16] $u_{b6} = R_{min}/R$
= 140 mm / 600 mm
= 0.23333
Segment 6: bend radius utilisation
[17] $u_{bend} = max\ R_{min}/R$
= 3 bend(s)
= 0.35
Bend radius utilisation (tightest bend)
[18] R_{bend} = min R of the bends
= 3 bend(s)
= 400 mm
Smallest bend radius of the route
[21] wg = w * 9.80665 m/s^2
= 1.18 kg/m × 9.80665 m/s^2
= 11.572 N/m
Weight per metre of the bundle on the containment
[23] $Ta_1 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 150 N + 11.572 N/m × 0.3 × 6 m
= 170.83 N
Segment 1 (as listed): tension after the straight length
[24] $Ta_2 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 170.83 N × cosh(0.47124) + √(170.83² + (11.572 N/m × 0.5 m)²) × sinh(0.47124)
= 273.71 N
Segment 2 (as listed): tension after the bend
[25] $Pa_2 = T_2/R$
= 273.71 N / 0.5 m
= 547.43 N/m
Segment 2 (as listed): sidewall pressure
[26] $Ta_3 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 273.71 N + 11.572 N/m × 0.3 × 14 m
= 322.31 N
Segment 3 (as listed): tension after the straight length
[27] $Ta_4 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 322.31 N × cosh(0.47124) + √(322.31² + (11.572 N/m × 0.4 m)²) × sinh(0.47124)
= 516.36 N
Segment 4 (as listed): tension after the bend
[28] $Pa_4 = T_2/R$
= 516.36 N / 0.4 m
= 1290.9 N/m
Segment 4 (as listed): sidewall pressure

[29] $Ta_5 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 516.36 N + 11.572 N/m × 0.3 × 9 m
= 547.6 N
Segment 5 (as listed): tension after the straight length
[30] $Ta_6 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 547.6 N × cosh(0.23562) + √(547.6² + (11.572 N/m × 0.6 m)²) × sinh(0.23562)
= 693.1 N
Segment 6 (as listed): tension after the bend
[31] $Pa_6 = T_2 / R$
= 693.1 N / 0.6 m
= 1155.2 N/m
Segment 6 (as listed): sidewall pressure
[32] $Tb_6 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 150 N × cosh(0.23562) + √(150² + (11.572 N/m × 0.6 m)²) × sinh(0.23562)
= 189.89 N
Segment 6 (reversed): tension after the bend
[33] $Pb_6 = T_2 / R$
= 189.89 N / 0.6 m
= 316.49 N/m
Segment 6 (reversed): sidewall pressure
[34] $Tb_5 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 189.89 N + 11.572 N/m × 0.3 × 9 m
= 221.14 N
Segment 5 (reversed): tension after the straight length
[35] $Tb_4 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 221.14 N × cosh(0.47124) + √(221.14² + (11.572 N/m × 0.4 m)²) × sinh(0.47124)
= 354.28 N
Segment 4 (reversed): tension after the bend
[36] $Pb_4 = T_2 / R$
= 354.28 N / 0.4 m
= 885.7 N/m
Segment 4 (reversed): sidewall pressure
[37] $Tb_3 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 354.28 N + 11.572 N/m × 0.3 × 14 m
= 402.88 N
Segment 3 (reversed): tension after the straight length
[38] $Tb_2 = T_1 \cdot \cosh(\mu_e \cdot \theta) + \sqrt{(T_1^2 + (w \cdot g \cdot w_c \cdot R)^2)} \cdot \sinh(\mu_e \cdot \theta)$
= 402.88 N × cosh(0.47124) + √(402.88² + (11.572 N/m × 0.5 m)²) × sinh(0.47124)
= 645.43 N
Segment 2 (reversed): tension after the bend
[39] $Pb_2 = T_2 / R$
= 645.43 N / 0.5 m
= 1290.9 N/m
Segment 2 (reversed): sidewall pressure
[40] $Tb_1 = T_1 + w \cdot g \cdot w_c \cdot \mu \cdot L$
= 645.43 N + 11.572 N/m × 0.3 × 6 m
= 666.26 N
Segment 1 (reversed): tension after the straight length
[41] $T_{max,a}$ = T at the end of the route, as listed
= 693.1 N
= 693.1 N
Maximum tension, pulling in the listed order

[42] $T_{max,b}$ = T at the end of the route, reversed
= 666.26 N
= 666.26 N
Maximum tension, pulling in the reverse order
[43] $P_{max,a}$ = max SWP, as listed
= 1290.9 N/m
= 1290.9 N/m
Maximum sidewall pressure, pulling in the listed order
[44] $P_{max,b}$ = max SWP, reversed
= 1290.9 N/m
= 1290.9 N/m
Maximum sidewall pressure, pulling in the reverse order
[45] $T_{cab} = N \cdot \min(T_{max,i})$
= 6 × 200 N
= 1200 N
Pulling tension limit of the bundle from the cables (equal load share, weakest cable)
[47] $max_{tension} = \min(T_{max,a} ; T_{max,b})$
= 693.1 N ; 666.26 N
= 666.26 N
Maximum pulling tension, recommended direction
[48] P_{max} = max SWP in the recommended direction
= 1290.9 N/m
= 1290.9 N/m
Maximum sidewall pressure, recommended direction
[49] $u_{Tc1} = (T/N)/(k_n \cdot T_{max,i})$
= 111.04 N / (1 × 600 N)
= 0.18507
Servo power: share of the pull against its own limit
[50] $u_{Tc2} = (T/N)/(k_n \cdot T_{max,i})$
= 111.04 N / (1 × 200 N)
= 0.55521
Encoder and control: share of the pull against its own limit
[51] $u_T = max_{tension} / T_{cab}$
= 666.26 N / 1200 N
= 0.55521
Tension utilisation against the cable limit (the weakest cable)
[52] $u_P = P_{max} / P_{lim}$
= 1290.9 N/m / 3000 N/m
= 0.43028
Sidewall pressure utilisation against the limit
[53] $f_{lim} = f_{max}$
= 0.5
= 50 %
Space factor limit as a percentage

5. Results

SYMBOL	VALUE	DESCRIPTION
fill	44.75 %	Fill (governing)
$max_{tension}$	666.3 N	Maximum tension, recommended direction
A_{cab}	562.3 mm²	Total cable cross-section
A_{int}	1256.6 mm²	Internal cross-section
OD_{max}	14 mm	Largest cable diameter

SYMBOL	VALUE	DESCRIPTION
w	1.180 kg/m	Mass per metre of the bundle
R_{min}	140 mm	Minimum bend radius
$T_{max,a}$	693.1 N	Maximum tension, as listed
$T_{max,b}$	666.3 N	Maximum tension, reversed
$P_{max,a}$	1291 N/m	Maximum sidewall pressure, as listed
$P_{max,b}$	1291 N/m	Maximum sidewall pressure, reversed
P_{max}	1291 N/m	Maximum sidewall pressure, recommended direction
T_{cab}	1200 N	Pulling tension limit from the cables
direction	"Reverse: pull from segment 6 to segment 1"	Recommended pull direction
u_{fill}	0.90	Fill utilisation
u_{bend}	0.35	Bend radius utilisation
u_T	0.56	Cable tension utilisation
u_P	0.43	Sidewall pressure utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Fill against the space factor limit	44.75 %	50 %	0.90	
PASS	Pulling tension against the cable limit	666.3 N	1200 N	0.56	
PASS	Sidewall pressure against the limit	1291 N/m	3000 N/m	0.43	
PASS	Smallest bend radius needed against the tightest bend	140 mm	400 mm	0.35	[1](reg. 522.8)
PASS	Largest cable against the smaller internal side	largest cable 14 mm	conduit internal diameter 40 mm	–	

7. Applicability

WARNING records: tabulated values not taken from a reference record are f_{max} (Containment space factor); n_{bend} (Minimum bend radius multiple); μ (Pulling friction coefficient); P_{lim} (Pulling tension, grip and sidewall pressure limits). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
$bend_{model}$: not entered, so the default of 'Horizontal bend, cable on the containment floor (weight term)' is used.

8. Verification

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Template elec.containment@1.3.0 Content hash f97b10382d6cbf9a Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. BS EN 61386-1:2008.
3. Beer and Johnston, Vector Mechanics for Engineers: Statics.